

Chapter P5: Radioactive materials

Overview

The terms 'radiation' and 'radioactivity' are often interchangeable in the public mind. Because of its invisibility, radiation is commonly feared. A more objective evaluation of risks and benefits is encouraged through developing an understanding of the many practical uses of radioactive materials.

Topic P5.1 begins by considering the evidence of a nuclear model of the atom, including Rutherford's alpha particle scattering experiment. It then uses the nuclear model to explain what happens during radioactive decay. The properties of alpha, beta and

gamma radiation are investigated and ideas about half-life are developed. In Topic P5.2 learners learn about the penetration properties of ionising radiation which leads to a consideration of the use of radioactive materials in the health sector, and how they can be handled safely. In the context of health risks associated with irradiation and/or contamination by radioactive material, they also learn about the interpretation of data on risk.

Topic P5.3 describes nuclear fission and nuclear fusion. Learners have the opportunity to learn more about the issues that surround decisions about the best way to generate electricity.

Learning about the radioactivity before GCSE (9–1)

There is no formal learning about radioactivity before Key Stage 4, but learners will have ideas about radioactivity, nuclear energy and radiation from everyday language. From Topic P1.2 learners should:

- recall that in each atom its electrons are arranged at different distances from the nucleus

- recall that gamma rays are emitted from the nuclei of atoms
- be able to describe how ionising radiation can have hazardous effects, notably on human bodily tissues.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about radioactivity at GCSE (9–1)

P5.1 What is radioactivity?	
Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
An atom has a nucleus, made of protons and neutrons, which is surrounded by electrons. The modern model of the atom developed over time as scientists rejected earlier models and proposed new ones to fit the currently available evidence. Each stage relied on scientists using reasoning to propose models which fitted the evidence available at the time. Models were rejected, modified and extended as new evidence became available (IaS3). After the discovery of the electron in the 19th century by Thomson scientists imagined that atoms were small particles of positive matter with the negative electrons spread through, like currants in a cake. This was the model used until 1910 when the results of the Rutherford-Geiger-Marsden alpha particle scattering experiment provided evidence that a gold atom contains a small, massive, positive region (the nucleus). Atoms are small – about 10^{-10} m across, and the nucleus is at the centre, about a hundred-thousandth of the diameter of the atom. Each atom has a nucleus at its centre and that nucleus is made of protons and neutrons. For an element, the number of the protons is always the same but the number of neutrons may differ. Forms of the same element with different numbers of neutrons are called the isotopes of the element. Interpreting the unexpected results of the Rutherford-Geiger-Marsden experiment required imagination to consider a new model of the atom.	1. describe the atom as a positively charged nucleus surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with almost all of the mass in the nucleus
	2. describe how and why the atomic model has changed over time to include the main ideas of Dalton, Thomson, Rutherford and Bohr
	3. recall the typical size (order of magnitude) of atoms and small molecules
	4. recall that atomic nuclei are composed of both protons and neutrons, and that the nucleus of each element has a characteristic positive charge
	5. recall that nuclei of the same element can differ in nuclear mass by having different numbers of neutrons, these are called isotopes
	6. use the conventional representation to show the differences between isotopes, including their identity, charge and mass
	7. recall that some nuclei are unstable and may emit alpha particles, beta particles, or neutrons, and electromagnetic radiation as gamma rays
	8. relate emissions of alpha particles, beta particles, or neutrons, and gamma rays to possible changes in the mass or the charge of the nucleus, or both

Linked learning opportunities

Specification links

- How has our understanding of the structure of atoms developed over time? (C2.1)

Ideas about Science

- Explain how the development of the nuclear model of the atom is an example of how scientific explanations become accepted (IaS3).

Practical work

- Collect data to calculate the half-life of a radioactive isotope.
- Use a random event such as dice-throwing to model radioactive decay.

P5.1 What is radioactivity?

Teaching and learning narrative

Some substances emit ionising radiation all the time and are called radioactive. The ionising radiation (alpha, beta, gamma, and neutron) is emitted from the unstable nucleus of the radioactive atoms, which as a result become more stable.

Alpha particles consist of two protons and two neutrons, and beta particles are identical to electrons. Gamma radiation is very high frequency electromagnetic radiation.

Radioactive decay is a random process. For each radioactive isotope there is a different constant chance that any nucleus will decay. Over time the activity of radioactive sources decreases, as the number of undecayed nuclei decreases.

The time taken for the activity to fall to half is called the half-life of the isotope and can be used to calculate the time it takes for a radioactive material to become relatively safe.

Assessable learning outcomes

Learners will be required to:

9. use names and symbols of common nuclei and particles to write balanced equations that represent the emission of alpha, beta, gamma, and neutron radiations during radioactive decay
M1b, M1c, M3c
10. explain the concept of half-life and how this is related to the random nature of radioactive decay
- 11. calculate the net decline, expressed as a ratio, in a radioactive emission after a given (integral) number of half-lives**
M1c, M3d
12. interpret activity-time graphs to find the half-life of radioactive materials
M1c, M2g, M4a, M4c

Linked learning opportunities

P5.2: How can radioactive materials be used safely?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Ionising radiation can damage living cells and these may be killed or may become cancerous, so radioactive materials must be handled with care. In particular, a radioactive material taken into the body (contamination) poses a higher risk than the same material outside as the material will continue to emit ionising radiation until it leaves the body. Whilst ionising radiation can cause cancer, it can also be used for imaging inside the body and to kill cancerous cells. Doctors and patients need to consider the risks and benefits when using ionising radiation to treat diseases.	1. recall the differences in the penetration properties of alpha particles, beta particles and gamma rays
	2. recall the differences between contamination and irradiation effects and compare the hazards associated with each of these
	3. describe the different uses of nuclear radiations for exploration of internal organs, and for control or destruction of unwanted tissue
	4. explain how ionising radiation can have hazardous effects, notably on human bodily tissues
	5. explain why the hazards associated with radioactive material differ according to the radiation emitted and the half-life involved

Linked learning opportunities

Specification links:

- What are the risks and benefits of using electromagnetic radiations? (P1.2)

Practical work

- Collect and interpret data to show the penetration properties of ionising radiations.

Ideas about Science

- Discuss ideas about correlation and cause in the context of links between ionising radiation and cancer (IaS3).
- Discuss the uses of ionising radiation, with reference to its risks and benefits (IaS4).

P5.3 How can radioactive materials be used to provide energy? (*separate science only*)

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Nuclear fuels are radioactive materials that release energy during changes in the nucleus. In nuclear fission a neutron splits a large and unstable nucleus (such as some isotopes of uranium and plutonium) into two smaller parts, roughly equal in size, releasing more neutrons, which may go on to make further collisions. Energy is released from the nucleus, carried away as kinetic energy of the particles and also by gamma radiation. This release of energy from the nuclear store is analogous to that released from the chemical store of explosives like TNT but it is considerably larger. If brought close enough together, hydrogen nuclei can fuse into helium nuclei, releasing energy, and this is called nuclear fusion. The demand for energy is continually increasing and nuclear fuels are an alternative energy source to fossil fuels. The risks and benefits need to be compared when making decisions about how to generate electricity.	1. recall that some nuclei are unstable and may split into two nuclei and that this is called nuclear fission
	2. relate the energy released during nuclear fission to the emission of ionising radiation and the kinetic energy of the resulting particles
	3. explain how nuclear fission can lead to further fission events in a chain reaction
	4. describe the process of nuclear fusion and recall that in this process some of the mass may be converted into the energy of radiation

Linked learning opportunities

Specification links:

- How should electricity be generated? (P3.2)

Ideas about Science

- Discuss the risks and benefits of generating electricity using nuclear fission. Suggest reasons why different decisions on the same issue might be appropriate in view of differences in personal, social, or economic context (IaS4).